SYNAPSE Survey Questionnaire: Clinic Module

For Clinic Staff

This survey questionnaire evaluates the Clinic Module of SYNAPSE using four software product quality characteristics defined under ISO/IEC 25010: Functional Suitability, Security, Reliability, and Interaction Capability. These characteristics were selected based on their direct relevance to a web-based clinic management module that handles sensitive student and employee health data, serves multiple user roles, and operates in a healthcare-adjacent environment. Completeness of function coverage and accuracy of data processing are primary indicators of a health information system's suitability for its intended operational context (Tummers et al., 2021). Systems operating in real-time clinical environments require high availability, fault-tolerant architectures, and defined recovery mechanisms to ensure continuity of care (Aceto et al., 2020). Usability evaluation of health information technology systems relies on consistent methods for assessing learnability, task efficiency, and error prevention as core indicators of system quality across diverse, multi-role user groups (Abareshi et al., 2025). Each characteristic is assessed through a set of targeted statements derived from its subcategories, which collectively measure whether the module meets its intended functional requirements, protects confidential health information, performs dependably under real-world conditions, and delivers a usable experience for clinic staff.

Role: _____ **Date:** _____

Directions: Please rate each statement by placing a check mark (✓) under the number that best matches your evaluation.

Rating Scale: 5 – Strongly Agree; 4 – Agree; 3 – Neutral; 2 – Disagree; 1 – Strongly Disagree

A. Functional Suitability

No.	Statement	5	4	3	2	1
1	**Functional Completeness:** The module covers all the required functions for clinic management, including patient records, consultations, referrals, medicine inventory, and reports.					
2	**Functional Completeness:** The module supports the Clinic Admin, Doctor, and Nurse roles with their corresponding permissions, ensuring that no required function is left inaccessible to its intended user.					
3	**Functional Completeness:** The module supports the Clinic Admin, Doctor, and Nurse roles with their corresponding permissions, ensuring that no required function is left inaccessible to its intended user.					

No.	Statement	5	4	3	2	1
4	**Functional Correctness:** The module accurately processes and stores patient records and attendance data.					
5	**Functional Correctness:** The module accurately manages medicine inventory and issues timely low-stock and expiration alerts.					
6	**Functional Correctness:** The module accurately retrieves and displays student information and referral history when a school ID is cross-checked by name against the scheduled appointment, and correctly handles referrals received from counselling and from employees.					
7	**Functional Appropriateness:** The module provides the right tools and features to support clinic staff in completing their tasks.					

B. Security

No.	Statement	5	4	3	2	1
1	**Confidentiality:** The system ensures that sensitive health data is accessible only to authorized users based on their assigned role.					
2	**Integrity:** The system prevents unauthorized modification of patient records, medicine inventory, referral statuses, and audit logs.					
3	**Non-repudiation:** The system maintains a verifiable record of all user actions, ensuring that transactions and data changes cannot be denied.					
4	**Accountability:** The system logs all authentication events, data access, referrals, and system activities with sufficient detail to trace actions back to individual users.					
5	**Authenticity:** The system verifies the identity of users through secure login credentials before granting access to any clinic data.					
6	**Resistance:** The system remains protected against unauthorized access attempts, including brute force					

No.	Statement	5	4	3	2	1
	login attempts, SQL injection, and other common attack methods.					

C. Reliability

No.	Statement	5	4	3	2	1
1	**Faultlessness:** The system performs its clinic functions without errors or failures under normal operating conditions.					
2	**Faultlessness:** The system correctly processes and updates referral statuses in real time, ensuring no referral is lost or left unacknowledged.					
3	**Availability:** The system remains accessible and operational during peak clinic hours without unplanned downtime.					
4	**Fault Tolerance:** The system continues to function or degrade gracefully when a component failure or network disruption occurs.					
5	**Recoverability:** The system restores full functionality and data integrity within an acceptable timeframe following an unexpected failure or outage.					

D. Interaction Capability

No.	Statement	5	4	3	2	1
1	**Appropriateness Recognizability:** The system clearly communicates its purpose and available functions so that clinic staff can immediately recognize its suitability for their role.					
2	**Learnability:** Clinic staff can learn to use the system effectively without extensive training.					
3	**Operability:** The system allows clinic staff to complete their tasks efficiently with minimal steps and without unnecessary complexity.					

No.	Statement	5	4	3	2	1
4	**Operability:** Users can access all integrated modules through a single login without needing to authenticate separately for each module.					
5	**User Error Protection:** The system prevents or warns users before committing actions that could result in data loss, incorrect entries, incorrect inventory changes, or unintended referrals.					
6	**Self-descriptiveness:** System messages and alerts (such as low stock, expiration, and referral updates) are clear and tell users what action to take.					

SYNAPSE Survey Questionnaire: Counselling Module

For Guidance Counsellors

This survey questionnaire evaluates the Counselling Module of SYNAPSE using four software product quality characteristics defined under ISO/IEC 25010: Functional Suitability, Security, Reliability, and Interaction Capability. These characteristics were selected based on their direct relevance to a web-based counselling management module that handles highly confidential student and employee information, serves multiple user roles, and operates in a healthcare-adjacent environment. Completeness of function coverage and accuracy of data processing are primary indicators of a health information system's suitability for its intended operational context (Tummers et al., 2021). Systems operating in real-time clinical environments require high availability, fault-tolerant architectures, and defined recovery mechanisms to ensure continuity of care (Aceto et al., 2020). Usability evaluation of health information technology systems relies on consistent methods for assessing learnability, task efficiency, and error prevention as core indicators of system quality across diverse, multi-role user groups (Abareshi et al., 2025). Each characteristic is assessed through a set of targeted statements derived from its subcategories, which collectively measure whether the module meets its intended functional requirements, protects confidential counselling information, performs dependably under real-world conditions, and delivers a usable experience for guidance counsellors.

Role: _____ **Date:** _____

Directions: Please rate each statement by placing a check mark (✓) under the number that best matches your evaluation.

Rating Scale: 5 – Strongly Agree; 4 – Agree; 3 – Neutral; 2 – Disagree; 1 – Strongly Disagree

A. Functional Suitability

No.	Statement	5	4	3	2	1
1	**Functional Completeness:** The module covers all the required functions for counselling management, including appointments, session records, referrals, and reports.					
2	**Functional Completeness:** The module supports the Guidance Counsellor role with its corresponding permissions, ensuring that no required function is left inaccessible to its intended user.					
3	**Functional Completeness:** The module provides an analytics dashboard that generates accurate and up-to-date counselling reports for administrative decision-making, with read-only access for Report Viewers.					

No.	Statement	5	4	3	2	1
4	**Functional Correctness:** The module processes counselling appointment bookings and rescheduling without errors, and correctly detects scheduling conflicts.					
5	**Functional Correctness:** The module accurately administers, processes, and stores standardized developmental assessments.					
6	**Functional Correctness:** The module accurately retrieves and displays student information and referral history when a school ID is cross-checked by name against the scheduled appointment, and correctly handles referrals received from the clinic and from employees.					
7	**Functional Appropriateness:** The module provides the right tools and features to support counsellors in completing their tasks.					

B. Security

No.	Statement	5	4	3	2	1
1	**Confidentiality:** The system ensures that sensitive counselling data is accessible only to authorized users based on their assigned role.					
2	**Integrity:** The system prevents unauthorized modification of counselling records, assessment results, referral statuses, and audit logs.					
3	**Non-repudiation:** The system maintains a verifiable record of all user actions, ensuring that transactions and data changes cannot be denied.					
4	**Accountability:** The system logs all authentication events, data access, referrals, and system activities with sufficient detail to trace actions back to individual users.					
5	**Authenticity:** The system verifies the identity of users through secure login credentials before granting access to any counselling data.					

No.	Statement	5	4	3	2	1
6	**Resistance:** The system remains protected against unauthorized access attempts, including brute force login attempts, SQL injection, and other common attack methods.					

C. Reliability

No.	Statement	5	4	3	2	1
1	**Faultlessness:** The system performs its counselling functions without errors or failures under normal operating conditions.					
2	**Faultlessness:** The system correctly processes and updates referral statuses in real time, ensuring no referral is lost or left unacknowledged.					
3	**Availability:** The system remains accessible and operational during peak counselling hours without unplanned downtime.					
4	**Fault Tolerance:** The system continues to function or degrade gracefully when a component failure or network disruption occurs.					
5	**Recoverability:** The system restores full functionality and data integrity within an acceptable timeframe following an unexpected failure or outage.					

D. Interaction Capability

No.	Statement	5	4	3	2	1
1	**Appropriate Recognizability:** The system clearly communicates its purpose and available functions so that counsellors can immediately recognize its suitability for their role.					
2	**Learnability:** Counsellors can learn to use the system effectively without extensive training.					
3	**Operability:** The system allows counsellors to complete their tasks efficiently with minimal steps and without unnecessary complexity.					

No.	Statement	5	4	3	2	1
4	**Operability:** Users can access all integrated modules through a single login without needing to authenticate separately for each module.					
5	**User Error Protection:** The system prevents or warns users before committing actions that could result in data loss, incorrect entries, double-booked appointments, or unintended referrals.					
6	**Self-descriptiveness:** System messages and alerts (such as appointment conflicts and referral updates) are clear and tell users what action to take.					

SYNAPSE Survey Questionnaire: Facilities/BMG Module

For BnG Staff

This survey questionnaire evaluates the Facilities/Biodegradable Waste Management (BMG) Module of SYNAPSE using four software product quality characteristics defined under ISO/IEC 25010: Functional Suitability, Security, Reliability, and Interaction Capability. These characteristics were selected based on their direct relevance to a web-based facilities module that records biodegradable waste batches and schedules and monitors an IoT-enabled mechanized BMG unit, serves multiple user roles, and involves physical equipment that must run dependably and safely. Each characteristic is assessed through a set of targeted statements derived from its subcategories, which collectively measure whether the module meets its intended functional requirements, protects batch records and unit controls from unauthorized changes, performs dependably under real-world conditions, and delivers a usable experience for facilities staff.

Role: _____ **Date:** _____

Directions: Please rate each statement by placing a check mark (✓) under the number that best matches your evaluation.

Rating Scale: 5 – Strongly Agree; 4 – Agree; 3 – Neutral; 2 – Disagree; 1 – Strongly Disagree

A. Functional Suitability

No.	Statement	5	4	3	2	1
1	**Functional Completeness:** The module covers all the required functions for BMG operations, including batch records, process logs, maintenance, archiving, and reports.					
2	**Functional Completeness:** The module supports the Facilities/Composting Staff role with its corresponding permissions, ensuring that no required function is left inaccessible to its intended user.					
3	**Functional Completeness:** The module provides an analytics dashboard that generates accurate and up-to-date BMG reports for administrative decision-making, with read-only access for Report Viewers.					
4	**Functional Correctness:** The module accurately records BMG batches (waste category, input and output weight, process logs, and status changes) and rejects entries where the recorded output exceeds the input.					

No.	Statement	5	4	3	2	1
5	**Functional Correctness:** The module correctly applies the schedule, speed (RPM), and strength settings to the mechanized BMG unit.					
6	**Functional Correctness:** The module accurately displays the unit's actual status, speed, and current load.					
7	**Functional Appropriateness:** The module provides the right tools and features to support facilities staff in completing their tasks.					

B. Security

No.	Statement	5	4	3	2	1
1	**Confidentiality:** The system ensures that BMG records are accessible only to authorized users based on their assigned role.					
2	**Integrity:** The system prevents unauthorized modification of BMG batch records, schedules, unit settings, and audit logs.					
3	**Non-repudiation:** The system maintains a verifiable record of all user actions, including changes to BMG schedules and settings, ensuring that changes cannot be denied.					
4	**Accountability:** The system logs all authentication events, BMG operations, and setting changes with sufficient detail to trace actions back to individual users.					
5	**Authenticity:** The system verifies the identity of users through secure login credentials before granting access to BMG data or to the controls of the mechanized BMG unit.					
6	**Resistance:** The system remains protected against unauthorized access attempts, including brute force login attempts, SQL injection, and other common attack methods.					

C. Reliability

No.	Statement	5	4	3	2	1
1	**Faultlessness:** The system performs its BMG functions without errors or failures under normal operating conditions.					
2	**Faultlessness:** The mechanized BMG unit runs its scheduled cycles consistently, without missed or duplicate starts.					
3	**Availability:** The BMG dashboard remains accessible and operational without unplanned downtime.					
4	**Fault Tolerance:** The mechanized BMG unit continues its stored schedule during a loss of network connection, and the system updates its status once the connection is restored.					
5	**Fault Tolerance:** The system detects motor overload or a jam in the mechanized BMG unit, stops the motor, and alerts the facilities staff.					
6	**Fault Tolerance:** The emergency stop and lid interlock stop the mechanized BMG unit immediately, even if the controller or the network fails.					
7	**Recoverability:** The system restores full functionality and data integrity within an acceptable timeframe following a failure or power loss, and the unit does not resume an interrupted cycle unsafely.					

D. Interaction Capability

No.	Statement	5	4	3	2	1
1	**Appropriateness Recognizability:** The system clearly communicates its purpose and available functions so that facilities staff can immediately recognize its suitability for their role.					
2	**Learnability:** Facilities staff can learn to use the system effectively without extensive training.					
3	**Operability:** Facilities staff can set the schedule, speed (RPM), and strength of the mechanized BMG					

No.	Statement	5	4	3	2	1
	unit and monitor its status through the dashboard with ease.					
4	**Operability:** Users can access all integrated modules through a single login without needing to authenticate separately for each module.					
5	**User Error Protection:** The system prevents or warns users before committing actions that could result in data loss, incorrect entries, cancelled batches, or out-of-range BMG settings.					
6	**Self-descriptiveness:** System messages and alerts (such as jam or overload warnings and schedule confirmations) are clear and tell users what action to take.					

SYNAPSE Survey Questionnaire: Personal Health Portal

For Students

This survey questionnaire evaluates the Personal Health Portal of SYNAPSE as experienced by students using four software product quality characteristics defined under ISO/IEC 25010: Functional Suitability, Security, Reliability, and Interaction Capability. These characteristics were selected based on their direct relevance to a web-based self-service portal that allows students to request clinic visits and guidance sessions, monitor appointment status, present a live queue ticket, and view their own health summary while protecting sensitive student health information. Completeness of function coverage and accuracy of data processing are primary indicators of a health information system's suitability for its intended operational context (Tummers et al., 2021). Systems operating in real-time clinical environments require high availability, fault-tolerant architectures, and defined recovery mechanisms to ensure continuity of care (Aceto et al., 2020). Usability evaluation of health information technology systems relies on consistent methods for assessing learnability, task efficiency, and error prevention as core indicators of system quality across diverse, multi-role user groups (Abareshi et al., 2025). Each characteristic is assessed through a set of targeted statements derived from its subcategories, which collectively measure whether the portal meets students' self-service needs, protects confidential health information, performs dependably under real-world conditions, and delivers a usable experience for students.

Role: _____ Date: _____

Directions: Please rate each statement by placing a check mark (✓) under the number that best matches your evaluation.

Rating Scale: 5 – Strongly Agree; 4 – Agree; 3 – Neutral; 2 – Disagree; 1 – Strongly Disagree

A. Functional Suitability

No.	Statement	5	4	3	2	1
1	**Functional Completeness:** The portal covers all the required self-service functions for students, including requesting clinic visits and guidance sessions, viewing appointment status, accessing the personal health summary, and receiving announcements, notifications, and surveys.					
2	**Functional Completeness:** The portal allows students to request both clinic and guidance appointments through one account without needing a separate login for each service.					
3	**Functional Completeness:** The portal accurately displays the student's profile, health summary (blood type and recorded allergies), and current waiting-room ticket and status after check-in.					

No.	Statement	5	4	3	2	1
4	**Functional Correctness:** The portal records the student's preferred date, preferred time, visit or session type, and reason accurately when an appointment is requested.					
5	**Functional Correctness:** The portal displays the correct appointment date, time, assigned staff, and status after the department confirms, reschedules, or completes the appointment.					
6	**Functional Correctness:** The portal accurately reflects the cancellation of a pending request or confirmed appointment and releases the reserved slot.					
7	**Functional Appropriateness:** The portal provides the tools students need to manage their own clinic and guidance appointments without contacting department staff.					

B. Security

No.	Statement	5	4	3	2	1
1	**Confidentiality:** The portal ensures that a student's health summary, appointment details, and any counselling information made available in the portal are accessible only to the student and to authorized clinic or guidance staff.					
2	**Integrity:** The portal prevents unauthorized modification of the student's profile, appointment requests, health summary, and counselling information.					
3	**Non-repudiation:** The portal maintains a verifiable record of appointment requests, cancellations, and profile changes, ensuring that these actions cannot be denied.					
4	**Accountability:** The portal logs authentication events and access to the student's personal health information with sufficient detail to trace actions back to individual users.					

No.	Statement	5	4	3	2	1
5	**Authenticity:** The portal verifies the student's identity through secure login credentials before granting access to personal health information.					
6	**Resistance:** The portal remains protected against unauthorized access attempts, including brute force login attempts, SQL injection, and other common attack methods.					

C. Reliability

No.	Statement	5	4	3	2	1
1	**Faultlessness:** The portal performs its self-service functions without errors or failures under normal operating conditions.					
2	**Faultlessness:** Appointment requests, cancellations, and status updates are processed correctly and reflected in the portal without duplicate or lost records.					
3	**Availability:** The portal remains accessible and operational during peak periods, such as enrollment, examination weeks, and clinic hours, without unplanned downtime.					
4	**Fault Tolerance:** The portal continues to function or degrade gracefully when a component failure or network disruption occurs, without exposing or corrupting student data.					
5	**Recoverability:** The portal restores full functionality and data integrity within an acceptable timeframe following an unexpected failure or outage.					

D. Interaction Capability

No.	Statement	5	4	3	2	1
1	**Appropriateness Recognizability:** The portal clearly communicates its purpose and available functions so that students can immediately recognize how to request a clinic visit or guidance session.					

No.	Statement	5	4	3	2	1
2	**Learnability:** Students can learn to use the portal effectively without extensive training or assistance.					
3	**Operability:** The portal allows students to complete appointment requests and cancellations efficiently with minimal steps and without unnecessary complexity.					
4	**Operability:** Students can access the portal and all integrated self-service functions through a single login without needing to authenticate separately for each service.					
5	**User Error Protection:** The portal prevents or warns students before committing actions that could result in a duplicate request, an accidental cancellation, or an incorrect appointment entry.					
6	**Self-descriptiveness:** Portal messages and status labels (such as “Awaiting confirmation,” “Confirmed,” “In Queue,” and “Cancelled”) are clear and tell students what action to take.					

SYNAPSE Survey Questionnaire: Personal Health Portal

For Employees

This survey questionnaire evaluates the Personal Health Portal of SYNAPSE as experienced by employees using four software product quality characteristics defined under ISO/IEC 25010: Functional Suitability, Security, Reliability, and Interaction Capability. These characteristics were selected based on their direct relevance to a web-based self-service portal that allows teaching and non-teaching employees to request clinic visits and guidance sessions, monitor appointment status, present a live queue ticket, and view their own health summary without exposing personal health information to supervisors or colleagues. Completeness of function coverage and accuracy of data processing are primary indicators of a health information system's suitability for its intended operational context (Tummers et al., 2021). Systems operating in real-time clinical environments require high availability, fault-tolerant architectures, and defined recovery mechanisms to ensure continuity of care (Aceto et al., 2020). Usability evaluation of health information technology systems relies on consistent methods for assessing learnability, task efficiency, and error prevention as core indicators of system quality across diverse, multi-role user groups (Abareshi et al., 2025). Each characteristic is assessed through a set of targeted statements derived from its subcategories, which collectively measure whether the portal meets employees' self-service needs, protects confidential health information, performs dependably under real-world conditions, and delivers a usable experience for employees.

Role: _____ Date: _____

Directions: Please rate each statement by placing a check mark (✓) under the number that best matches your evaluation.

Rating Scale: 5 – Strongly Agree; 4 – Agree; 3 – Neutral; 2 – Disagree; 1 – Strongly Disagree

A. Functional Suitability

No.	Statement	5	4	3	2	1
1	**Functional Completeness:** The portal covers all the required self-service functions for employees, including requesting clinic visits and guidance sessions, viewing appointment status, accessing the personal health summary, and receiving announcements, notifications, and surveys.					
2	**Functional Completeness:** The portal allows teaching and non-teaching employees to request both clinic and guidance appointments through one account without needing a separate login for each service.					

No.	Statement	5	4	3	2	1
3	**Functional Completeness:** The portal accurately displays the employee's profile (department, position, employment status, and teaching or non-teaching classification), health summary, and current waiting-room ticket after check-in.					
4	**Functional Correctness:** The portal records the employee's preferred date, preferred time, visit or session type, and reason accurately when an appointment is requested.					
5	**Functional Correctness:** The portal displays the correct appointment date, time, assigned staff, and status after the department confirms, reschedules, or completes the appointment.					
6	**Functional Correctness:** The portal accurately reflects the cancellation of a pending request or confirmed appointment and releases the reserved slot.					
7	**Functional Appropriateness:** The portal provides the tools employees need to manage their own clinic and guidance appointments around their work schedule without contacting department staff.					

B. Security

No.	Statement	5	4	3	2	1
1	**Confidentiality:** The portal ensures that an employee's health summary, appointment details, and any counselling information made available in the portal are accessible only to the employee and to authorized clinic or guidance staff, and not to supervisors or colleagues.					
2	**Integrity:** The portal prevents unauthorized modification of the employee's profile, appointment requests, health summary, and counselling information.					

No.	Statement	5	4	3	2	1
3	**Non-repudiation:** The portal maintains a verifiable record of appointment requests, cancellations, and profile changes, ensuring that these actions cannot be denied.					
4	**Accountability:** The portal logs authentication events and access to the employee's personal health information with sufficient detail to trace actions back to individual users.					
5	**Authenticity:** The portal verifies the employee's identity through secure login credentials before granting access to personal health information.					
6	**Resistance:** The portal remains protected against unauthorized access attempts, including brute force login attempts, SQL injection, and other common attack methods.					

C. Reliability

No.	Statement	5	4	3	2	1
1	**Faultlessness:** The portal performs its self-service functions without errors or failures under normal operating conditions.					
2	**Faultlessness:** Appointment requests, cancellations, and status updates are processed correctly and reflected in the portal without duplicate or lost records.					
3	**Availability:** The portal remains accessible and operational during peak periods, such as registration, grading periods, and clinic hours, without unplanned downtime.					
4	**Fault Tolerance:** The portal continues to function or degrade gracefully when a component failure or network disruption occurs, without exposing or corrupting employee data.					
5	**Recoverability:** The portal restores full functionality and data integrity within an acceptable timeframe following an unexpected failure or outage.					

D. Interaction Capability

No.	Statement	5	4	3	2	1
1	**Appropriateness Recognizability:** The portal clearly communicates its purpose and available functions so that employees can immediately recognize how to request a clinic visit or guidance session.					
2	**Learnability:** Employees, including those with limited technical experience, can learn to use the portal effectively without extensive training or assistance.					
3	**Operability:** The portal allows employees to complete appointment requests and cancellations efficiently around their work schedule with minimal steps and without unnecessary complexity.					
4	**Operability:** Employees can access the portal and all integrated self-service functions through a single login without needing to authenticate separately for each service.					
5	**User Error Protection:** The portal prevents or warns employees before committing actions that could result in a duplicate request, an accidental cancellation, or an incorrect appointment entry.					
6	**Self-descriptiveness:** Portal messages and status labels (such as “Awaiting confirmation,” “Confirmed,” “In Queue,” and “Cancelled”) are clear and tell employees what action to take.					